

Visa: A Verification Layer for Legal AI Agents

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Working paper. Results marked [PENDING — results being computed] were still running when this version was written.

Abstract

Legal AI tools now draft memos, compare contracts and review data rooms, but lawyers cannot rely on their output: tools invent authorities, attribute to real sources statements those sources do not make, and silently omit items. We ask whether a **verification layer**, a method that forces every claim to be tied to a verbatim, machine-checked source and every requirement of a reference document to be accounted for, makes legal AI agents more reliable. We present Visa, which applies this principle in two forms: an after-the-fact checker that verifies each claim of an AI answer against official French sources (Légifrance, Judilibre) under the rule *no proof, no green*; and a “concordance” method that an agent follows while working. We evaluate the method on M&A tasks from Harvey’s public Legal Agent Bench (LAB), with three agents (Legora, Claude Code with Sonnet 5.5, Mistral Medium 3.5), each run without and with the method on identical documents and instructions, graded against the official rubric by two judges from different model families. On the first development task (38 criteria), the method takes Legora from 26/38 to 38/38, a full task pass, by recovering four omitted closing deliverables; it moves Mistral Medium 3.5 from 23/38 to 25/38; it costs Claude Code one criterion (38/38 to 37/38), on a task Claude Code already solved. The two judges agree on every criterion of the four runs they both graded. Measurements on seven held-out test tasks and on a French citation test set are in progress.

1. Introduction

Trust is the main barrier to legal AI. In a 2025 survey of 4,457 French lawyers by the Conseil national des barreaux (CNB), 64% use ChatGPT, 46% cite errors as their main concern, and 70% of non-users say they do not trust these tools. The concern is concrete: Damien Charlotin’s database lists 2,145 court decisions dealing with hallucinated material, 14 of them in France; on 29 December 2025, the administrative court of Orléans identified 17 fictitious references in a single filing. Professional tools are not immune: Stanford researchers

measured incorrect or misleading answers in 17% of Lexis+ AI responses and 33% of Westlaw AI responses (Magesh et al., 2024). Since March 2026, the CNB’s guide on ethics and AI asks lawyers to document their use of AI, yet no tool produces that record.

Agentic benchmarks show the same pattern at the level of complete work products. On Harvey’s Legal Agent Bench (LAB), models pass roughly 90 to 96% of rubric criteria individually but rarely a whole task, because a task only counts if every criterion passes. On the Vals AI leaderboard of 1 October 2026, the best model completes 25.4% of tasks, Claude Fable 5 11.3%, and Mistral Medium 3.5 0.4%. Harvey attributes most failures to incomplete search and single omissions, and reports that “verify then correct” is the strongest harness practice it measured (+1.5 points).

Our premise is that verification should be a product layer rather than a property hoped for in each model. **Contributions:**

- **Visa**, a verification layer that checks each claim of a legal AI answer for existence, temporal validity at the date of the facts, normative rank, and support by the cited text, with a verbatim guard that prevents the verifier itself from hallucinating, and that renders the result as a map of evidence.
- **A concordance method** packaged as an agent skill: a forward pass over every requirement of the reference document, a reverse pass, verbatim quotes checked by script, and a report drafted from the ledger.
- **A controlled evaluation protocol** on public LAB tasks: pre-registered dev/test split, identical inputs across conditions, isolation of agents from rubrics and from each other, and two judges from different model families.
- **First results** across three agents, including one commercial legal AI product (Legora).

2. Related work

Legal reasoning benchmarks. LegalBench (Stanford, 2023) gathers 162 legal reasoning tasks under US law; it is now saturated, with the top eight models within 2.5 points. It measures answers, not verifiability.

Hallucination in legal tools. Magesh et al. (2024), “Hallucination-Free?”, is the reference measurement for commercial legal research tools; it attributes errors to retrieval of the wrong material (20 to 47% of incorrect answers), inapplicable sources such as the wrong jurisdiction or overruled authority (23 to 38%), and real sources misgrounded (28 to 61%).

LexAgentHallu (arXiv 2609.09754, September 2026) classifies 3,414 cases of legal agent hallucination into 27 types and documents “right answer, wrong reason”. A 2026 ICML paper (arXiv 2609.17546) describes legal source verification, including temporal validity and normative hierarchy, as an open problem.

Agentic work-product benchmarks. Harvey’s LAB (May 2026, MIT license) releases tasks with client documents, instructions and lawyer-written rubrics, plus an execution harness; our checkout contains 2,010 public tasks, 136 of them in M&A. The Vals AI leaderboard reports results on held-out tasks. Harvey also reports, on M&A due diligence over full data rooms, that agents read 0.1 to 0.5% of the documents.

Citation checking. Existing citation-checking products (Clearbrief, KeyCite) target US law; we found none that checks validity at the date of the facts or the rank of the norm under French law. ContractNLI (Koreeda and Manning, 2021), which labels contract hypotheses as entailed, contradicted or not mentioned with evidence spans, is a natural benchmark for Visa’s support check *[reference not yet verified against the primary source]*.

3. System

3.1 THREE FAILURE MODES, FOUR CHECKS

Visa targets three failure modes of legal AI output: a **fabricated or inapplicable source** (nonexistent, repealed, of the wrong rank), a **misgrounded claim** (a real source made to say something it does not say), and an **omission** (a requirement of the governing document that the output never mentions). An answer is first split into claims, each with the sources it cites. Each cited source then goes through four checks:

1. **Existence.** The source is retrieved from Légifrance (codes, statutes, decrees) or Judilibre (Cour de cassation decisions) through the official PISTE API, or from the case file for exhibits. Not found is red; an unidentifiable reference (“settled case law”) is grey.
2. **Validity at the date of the facts.** For codified articles, Visa retrieves every historical version and selects the one in force at the date of the facts. No applicable version is red; a version that differs from today’s text is orange, and the two texts are shown side by side.
3. **Rank.** A simplified hierarchy of norms (constitution, treaties, statutes, decrees, orders, circulars, case law). A circular invoked as binding is orange.
4. **Support.** An adversarial judge, prompted as opposing counsel, decides whether the applicable text supports the claim (supports, partial, does not support, off-topic). It must quote a passage of about ten words or more, and a script checks that this quote appears verbatim in the official text after normalization. A verdict whose quote cannot be found is discarded and the claim turns grey.

The claim takes the worst status of its sources. The governing rule is **no proof, no green**: an API failure, a vague citation or an unverifiable quote yields grey, never green.

3.2 TWO FORMS

After the fact. A lawyer pastes an AI-written memo and the date of the facts. Visa returns a map of evidence: claims on one side, official sources on the other, one colored link per verification, and, on click, the controls, the adversarial reasoning and the highlighted passage. In the web application, extraction and judging use Mistral models; in the Claude Code skill, the agent running the skill extracts and judges, while scripts perform retrieval, the date and rank checks, the verbatim guard and the rendering. Both share one rule module, so they return the same verdicts.

During the work. The same principle, packaged as a skill for agents that compare documents (“cross-document review”). The agent must:

1. assign roles: the reference document (what governs), the subject (what is checked), context documents;
2. **forward pass**: walk the reference section by section, including definitions and schedules, and write one ledger row per operative requirement, with a verbatim quote and its location;
3. find each row’s counterpart in the subject and label it *match*, *deviation* or *missing*;
4. **reverse pass**: add rows for subject content with no basis in the reference;
5. pass a **scripted gate** that checks every quote verbatim, every row’s completeness, every document’s use, and lists figures no row covers yet;
6. draft the report **from the ledger**, then re-check the report against the ledger before delivery.

For tools that do not run scripts (Legora), we paste the same method as text after the task instruction (Appendix A.3).

4. Experimental setup

Tasks. We use M&A tasks from the public LAB release that require cross-referencing documents. Before running any of them, we fixed a split in `bench/split-ma.json` (seed 20261004): three **development** tasks, on which we read rubrics and failures and may adjust the method, and seven **test** tasks, whose rubrics are never read and which are measured once. This paper reports development results only; test results are pending.

Split	Task	Documents
dev	compare-closing-checklist-against-ma-agreement	2
dev	review-disclosure-schedules-against-representations-for-completeness	5
dev	compare-closing-docs	12
test	7 tasks (change of control x2, disclosure schedule issues, data room red flags, third-party consents, representations vs. diligence, disclosure schedule markup)	5 to 19

Agents.

- **Legora** (web product). Documents are uploaded and the task instruction is pasted verbatim. Legora proposes a plan, which we approve unchanged. The deliverable is downloaded and graded.
- **Claude Code with Sonnet 5.5** (subscription). Each run happens in a fresh workspace containing only the documents and the instruction, with an adaptation of Harvey’s system preamble and Harvey’s file-format skill manuals. Personal instructions, memory, plugins, MCP servers and web tools are disabled, and the loaded configuration is recorded for each run.
- **Mistral Medium 3.5** (API, reasoning effort “high” requested), run in Harvey’s official harness: a network-less container with Harvey’s seven tools.

Conditions. Each agent runs each task twice, *without* and *with* the method, on identical documents and instructions. For Claude Code and Mistral, *with* adds the skill manual and its scripts, as Harvey’s harness does for skills; for Legora, it appends the method text.

Grading. We use the official rubric and the official per-criterion judge prompt (`rubric_criterion`). Two judges grade independently: Claude Opus 5.5 at maximum reasoning effort and GPT-5.5 at high effort. Both run through subscription command-line clients (Claude Code, Codex) without tools. A task passes only if all criteria pass.

Safeguards.

- Agents never see `task.json`, which holds the rubric. A run whose tool calls `touch task.json`, the benchmark checkout or another run’s workspace is flagged and excluded.
- Workspaces are archived and made unreadable once a run ends.
- During the day we found three isolation leaks and closed each: a shared `/tmp`, workspaces listable by their siblings, and large tool outputs stored outside the workspace. We audited every kept run’s transcript and found no run that read another run’s files or any rubric.

- One early run was lost to an archiving error and re-run; one exploratory run that loaded personal settings is excluded.

Differences from the official leaderboard.

- We use public tasks; Vals uses held-out tasks.
- Claude runs in Claude Code rather than in Harvey’s harness, so tools and system prompt differ.
- Our judges are Opus 5.5 and GPT-5.5 through subscriptions; Vals uses GPT-5.5 and Claude Sonnet 4.6 through APIs.
- Our results are therefore comparable across our conditions, not to leaderboard figures.

5. Results

5.1 DEVELOPMENT TASK 1: CLOSING CHECKLIST VS. ASSET PURCHASE AGREEMENT (38 CRITERIA)

Agent	Without method	With method	Judges
Legora	26/38 (fail)	38/38 (pass)	Opus 5.5 and GPT-5.5, identical verdicts
Claude Code, Sonnet 5.5	38/38 (pass)	37/38 (fail)	Opus 5.5 and GPT-5.5, identical verdicts
Mistral Medium 3.5	23/38 (fail)	25/38 (fail)	Claude Sonnet 5.5 only; dual grading pending

Legora. Without the method, Legora finds every value discrepancy (escrow amount, indemnification cap, escrow period, transition services fee, non-compete term, outside date, and an internal inconsistency in the buyer’s name) and cites its sources. It misses all four closing deliverables that the agreement requires and the checklist omits: the FIRPTA certificate, the Northwind Aerospace consent, the title insurance commitment and the landlord estoppel certificate. These four omissions account for 11 of its 12 failed criteria, because each one is graded separately for identification, section reference, recommendation and, for FIRPTA, severity. The twelfth is the closing-payment reconciliation. With the method, the same tool recovers all four omissions and passes all 38 criteria. It does so even though it inverted the roles, taking the checklist as reference; we infer that its reverse pass is what surfaced the missing deliverables.

Claude Code. Claude Code already solves this task without the method. With the method, it misses one criterion (C-038): the report states the correct closing cash payment but never shows the reconciliation ($\$87.5\text{M} - \$5.25\text{M escrow} - \$3.5\text{M holdback} = \78.75M). Both judges agree. The method's run was slightly cheaper (16 turns, 232 s, $\approx \$0.89$ API-equivalent, against 22 turns, 269 s, $\approx \$0.98$).

Mistral Medium 3.5. The method adds two criteria (23 to 25) without producing a pass. Mistral followed the method's steps (54 turns against 20), and the run took 24 minutes against 5, largely because of API rate limits.

Judge agreement. On the four runs graded by both judges (152 criterion verdicts), Opus 5.5 and GPT-5.5 agree on every criterion, including the identity of each failed criterion. On this task, a Claude judge does not grade Claude's output more leniently than a GPT judge does.

5.2 DEVELOPMENT TASKS 2 AND 3

[PENDING — results being computed]

Agent	Task 2: disclosure schedules vs. reps (without / with)	Task 3: closing documents (without / with)
Legora	– / –	– / –
Claude Code, Sonnet 5.5	– / –	– / –
Mistral Medium 3.5	– / –	– / –

5.3 HELD-OUT TEST TASKS (7 TASKS, MEASURED ONCE)

[PENDING — results being computed]

Agent	Tasks passed without	Tasks passed with	Criteria passed without	Criteria passed with
Claude Code, Sonnet 5.5	– / 7	– / 7	–	–
Legora	– / 7	– / 7	–	–
Mistral Medium 3.5	– / 7	– / 7	–	–

5.4 FRENCH CITATION TEST SET (AFTER-THE-FACT VISA)

[PENDING — results being computed]

A set of 40 to 60 short claims in French law, grouped into memos, each labelled green, orange, red or grey after verification on Légifrance and Judilibre. We will report the share of false claims detected (not green), false greens, false reds, and accuracy per error type: invented decision, nonexistent article, not in force at the date of the facts, misgrounded, circular, vague reference.

Measure	Value
False claims detected (not green)	—
False greens	—
False reds	—

6. Discussion

Why 38/38 here while the leaderboard reports 10 to 20%. LAB scores are all-or-nothing per task. If an agent passes each criterion with probability 0.95 independently, it passes all 38 criteria of task 1 with probability $0.95^{38} \approx 0.14$, which is the order of magnitude of the leaderboard. On a due diligence task with 500 criteria, the same agent would almost never pass ($0.95^{500} \approx 7 \times 10^{-12}$). A single passed task is therefore consistent with a low aggregate pass rate. Task 1 is also short (two documents). Our harness for Claude (Claude Code) and our judges differ from the leaderboard’s (Section 4). Public tasks may have been seen in training, which we cannot verify.

What the method contributes. On task 1, the failures the method removes are **omissions**, not misread values: the baseline Legora report is accurate on every discrepancy it reports. This is the failure mode Harvey identifies as dominant, and the one that all-or-nothing scoring punishes most, since one missing deliverable costs three or four criteria. The forward pass turns “find the deviations” into “account for every requirement”, and the reverse pass catches what the forward pass framing misses.

When it does not help. When the agent already covers the task, the method adds constraints without adding coverage. Claude Code lost the one criterion that required showing arithmetic, a step the skill asks for but does not check by script. This suggests making every computed figure in the report pass the same scripted gate as quotes. Any such change must be validated on development tasks only.

7. Limitations

- **Small n and a single run per condition.** Agent runs vary; one task cannot establish an effect size.
- **Development tasks only so far.** Task 1 rubrics were read before the Legora method text was pasted. The text is the generic method from the skill, written earlier, and was not tailored to the four omissions, but only the held-out results can show the method generalizes.
- **Non-official judges.** The judges run through subscription clients that do not expose temperature; Mistral's runs on task 1 were graded by one judge only.
- **Public tasks.** Possible training contamination.
- **Human steps in Legora.** Plan approval is manual, although identical across conditions. Legora inverted reference and subject in the method condition.
- **Different harnesses.** Claude Code and Legora are not run in Harvey's harness, so cross-agent comparisons mix model and tool effects; within-agent comparisons (without vs. with) do not.

8. Conclusion and future work

A verification layer is cheap to add and agent-agnostic. On the first M&A task it turned a failing commercial tool into a passing one by forcing it to account for every requirement of the governing document. Next steps:

- complete the held-out measurement and the French citation test set;
- time a lawyer verifying the demo memo by hand versus with the evidence map;
- grade with the official API judges;
- extend the method to due diligence over full data rooms (2,600 to 4,000 documents per LAB task), adding an inventory of the data room (duplicates, unsigned drafts, amendments linked to their contracts), a coverage register of what was read, set aside (with a reason) or unreadable, and a verified extract for each red flag.

Appendix A. Reproducibility

All code is in `github.com/cedric190703/harness-law`, branch `youssef/bench-revue`. The LAB checkout lives in `harvey-labs/` (not committed).

A.1 COMMANDS

```
# Harvey's harness (Mistral), without and with the skill, graded by two judges
python3 bench/run.py --model mistral-medium-3.5 --effort high \
  --tasks <task> --judges "claude-code-opus-5-5@max" "codex-gpt-5.5@high"

# Claude Code through the subscription, isolated workspace, grade later
python3 bench/externe.py claude-code <task> --model sonnet [--skill] --pas-
noter

# A deliverable produced in another tool (Legora)
python3 bench/externe.py preparer <task>
python3 bench/externe.py importer <task> --source legora --fichier <file> [--
condition skill]

# (Re)grade existing runs with both judges
python3 bench/externe.py noter <run_id> --judges "claude-code-opus-5-5@max"
"codex-gpt-5.5@high"

# Review pages and dashboard
python3 bench/revue.py <task>
python3 bench/tableau.py

# Visa application: lint, types, tests
cd visa/app && bun run check
```

A.2 RUNS REPORTED IN SECTION 5.1

Agent	Condition	Run identifier (under <code>harvey-labs/results/corporate-ma/compare-closing-checklist-against-ma-agreement/</code>)
Legora	without	<code>legora-base/20261004-131842-558458</code>
Legora	with	<code>legora-skill/20261004-133520-495811</code>
Claude Code	without	<code>claude-code-sonnet-base/20261004-131336-947962</code>
Claude Code	with	<code>claude-code-sonnet-skill/20261004-131831-582658</code>
Mistral	without	<code>mistral-medium-3.5-high-base/20261004-132603-069767</code>
Mistral	with	<code>mistral-medium-3.5-high-skill/20261004-132603-069885</code>

A.3 METHOD TEXT PASTED INTO LEGORA (TASK 1)

Compare the attached APA against the closing checklist and produce a categorized, severity-rated deviation report. Output: `closing-checklist-deviation-report.docx`.

Use this verification method (concordance table). One missed or misstated item makes the report wrong, so trade speed for completeness:

1. Roles: identify the REFERENCE document (what governs), the SUBJECT document (what is checked) and any context documents. Read every document in full.
2. Forward pass: build a concordance table (use a Tabular Review if helpful). Walk the REFERENCE section by section, including definitions, schedules and exhibits. Every operative requirement gets a row (term, amount, date, party, condition, deliverable, certificate, consent, schedule), not only the ones that look important. For each row: reference section + verbatim quote, the SUBJECT counterpart + verbatim quote, and a status: match / deviation / missing (in the reference, absent from the subject).
3. Reverse pass: walk the SUBJECT section by section and add a row for anything with no basis in the reference (added), including wrong cross-references.
4. Omissions are deviations. Treat every “missing” row as a finding.
5. For every non-match row: both exact values, the concrete impact (quantified when possible, show the arithmetic), severity (Critical / High / Medium / Low) and a specific recommendation citing the section.
6. Draft the report from the table, most severe first, with a dedicated section for items required by the reference but missing from the subject.
7. Before finishing, check the report against the table: every deviation, missing and added row must appear in the report with its section reference. Fix and re-check until it does.

In the Legora interface, the same text was typed as a single paragraph with the steps numbered (1) to (7).

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